

Variables

A **variable** is a **named memory location used to store a value** that can change during program execution.

Types of C Variables

An integer variable can hold only an integer constant, a real variable can hold only a real constant and a character variable can hold only a character constant.

Rules for Constructing Variable Names

- (a) A variable name is any combination alphabets, digits or underscores
- (b) The first character in the variable name must be an alphabet or underscore.
- (c) No commas or blanks are allowed within a variable name
- (d) No special symbol other than an underscore can be used in a variable name.

si_int
m_hra
pop_e_89
_a
—

Variable Declaration

Before using a variable, we normally declare it by specifying its **data type and name**.

```
int age;  
float salary;  
char grade;
```

Variable Initialization

Assigning an initial value while declaring the variable is called **initialization**

```
int age = 20;  
float salary = 25000.50;  
char grade = 'A';
```

Assigning a New Value

The value of a variable can be changed during execution.

```
int age = 20;  
age = 25;
```

```
#include <stdio.h>  
int main()  
{  
    int age = 20;
```

```

float marks = 85.5;
char grade = 'A';
printf("Age  = %d\n", age);
printf("Marks = %.2f\n", marks);
printf("Grade = %c\n", grade);
return 0;
}
Age  = 20
Marks = 85.50
Grade = A

```

Reserved words

Reserved words (also called **keywords**) are words that have a **special meaning in the C language**.

They are **predefined by the C language**, so we **cannot use them as variable names, function names, or other user-defined identifiers**.

Common C Reserved Words

Reserved Word	Purpose
auto	Automatic storage class
break	Exit from loop or switch
case	Case in switch
char	Character data type
const	Constant value
continue	Skip current loop iteration
default	Default case in switch
do	Do-while loop
double	Double-precision floating-point type
else	Alternative condition
enum	Enumeration
extern	External variable/function declaration
float	Floating-point type
for	For loop
goto	Jump to a labeled statement
if	Conditional statement
int	Integer data type
long	Long integer type
register	Register storage class
return	Return from a function
short	Short integer type
signed	Signed data type
sizeof	Finds size of a data type/variable
static	Static storage class
struct	Structure
switch	Multiple-choice selection
typedef	Creates a type alias

union	Union
unsigned	Unsigned data type
void	No value/type
volatile	Prevents certain compiler optimizations
while	While loop